DATAWISE AI

DataWise AI: Complete Documentation

Comprehensive System Reference, Architecture Manual, and Operating Handbook

Document Type:	Complete Technical Documentation Manual
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Orchestration:	LangGraph 16-Agent Autonomous Architecture

Architectural Foundation: Deterministic ML execution engine paired with autonomous multi-agent reasoning, strict data leakage boundaries, and human-in-the-loop validation.

Table of Contents & Platform Overview

This master documentation manual consolidates the entirety of the DataWise AI platform specifications, covering foundational concepts, multi-agent graph topologies, database schemas, API surfaces, operational procedures, and maintenance protocols.

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02 System Architecture	Frontend, backend, database, agent orchestrator, storage	docs/02_system_architecture/
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09 Developer Guide	Local setup, extending agents, adding ML algorithms	docs/09_developer_documentation/
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1. Architectural Foundations

DataWise AI was built to solve the fragility of existing AI assistants in data science. Unlike general-purpose chatbots that generate unvalidated Python snippets or guess statistical metrics, DataWise AI operates as a controlled, deterministic state machine where every action is logged, reversible, and backed by verifiable scikit-learn code execution.

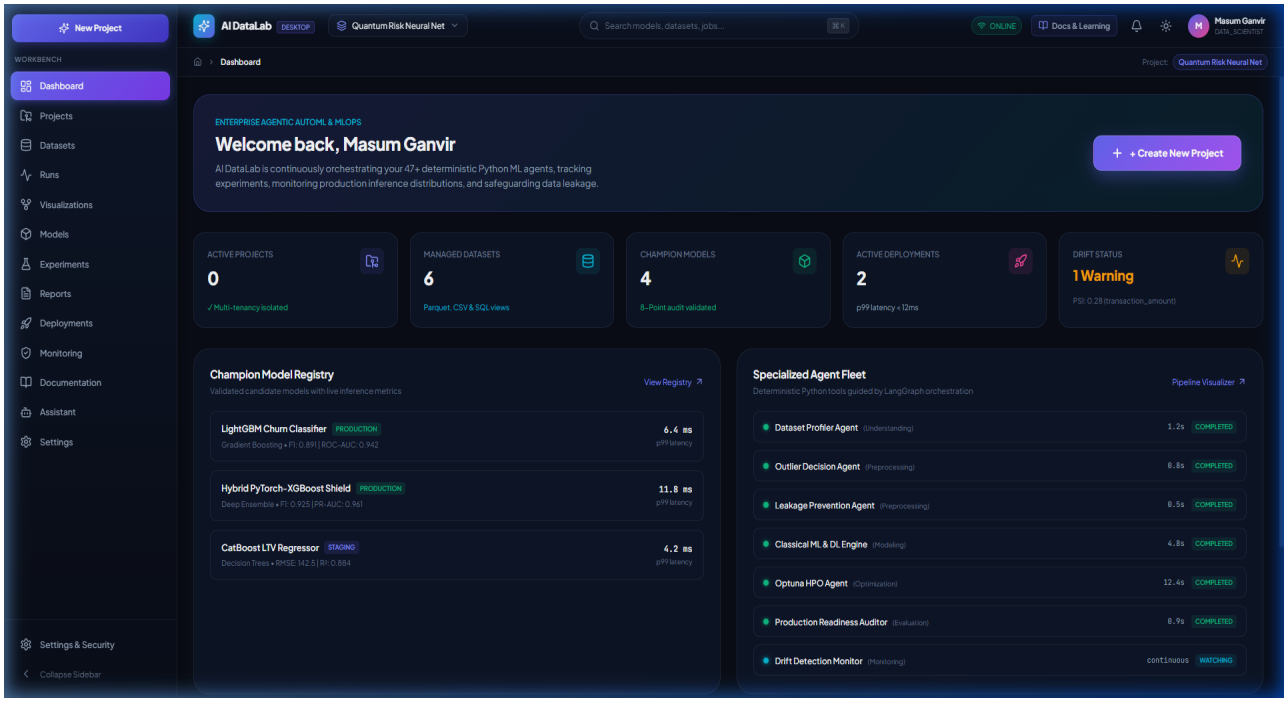


Figure: DataWise AI Executive Project Dashboard & Experiment Registry

2. State Management & LangGraph Execution Loop

The backbone of DataWise AI is `DataScienceState`, a Pydantic/TypedDict state container that persists across all agent steps. The state records the raw dataset path, processed dataframe fingerprints, cleaning recommendations, active target column, selected feature sets, candidate model rankings, Optuna trial histories, and human approval flags.

State Field	Data Type	Role in Multi-Agent Execution
Field Name	Type	Description & Usage in Graph
project_id	str (UUID)	Unique identifier linking all executions and persisted artifacts
dataset_profile	dict	Output of DatasetProfilingAgent (shapes, dtypes, nulls, cardinality)
data_quality_report	dict	Issues detected: missing values, duplicates, outliers, class skew
problem_definition	dict	Structured target, task type, business objectives, constraints
approved_steps	list[str]	List of transformation steps explicitly authorized by the user
fitted_pipeline_path	str	Filesystem/S3 path to the serialized scikit-learn ColumnTransformer
model_leaderboard	list[dict]	Ranked candidate models with cross-validation and test scores
error_log	list[dict]	Structured history of exceptions and autonomous recovery attempts

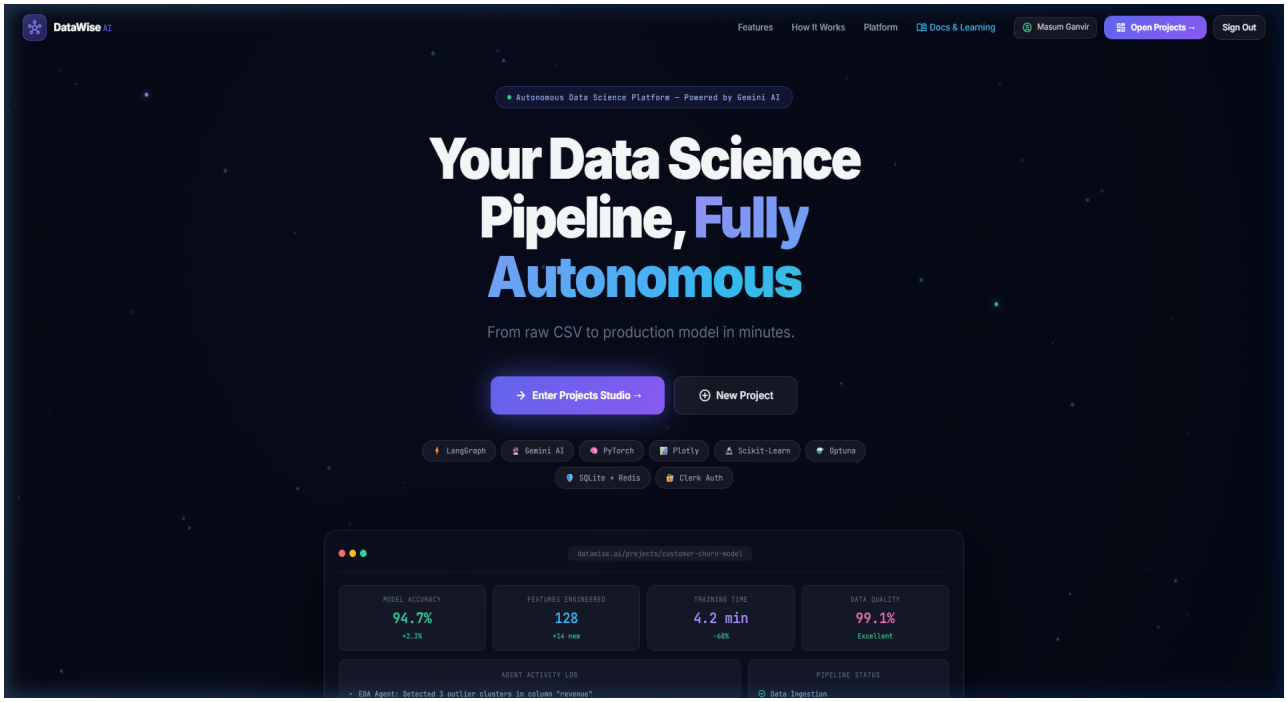


Figure: DataWise AI Authentication & Secure Gateway